

Hank Chen

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EDUCATION

University of Illinois Urbana-Champaign

Jan 2025 – Dec 2026

Master of Computer Science; GPA: 4.0/4.0

Champaign, IL

- Relevant Coursework: Deep Learning for Computer Vision, Distributed Systems, Database Systems

National Cheng Kung University

Sept 2019 – June 2023

Bachelor of Science in Computer Science and Statistics; GPA: 3.99/4.3

Tainan, Taiwan

WORK EXPERIENCE

Taiwan Semiconductor Manufacturing Company

June 2025 – Aug 2025

Software Engineer Intern | Python, React, Azure, Docker, Kubernetes, Helm, FastAPI

Hsinchu, Taiwan

- Pioneered a platform for organizational change with status-update portals and management dashboards in **15** fabs
- Cut **80%** tracking progress time by automating Teams reminders systems with **20+** modular Python utilities
- Lowered **90%** code change time by migrating Delphi systems to Python/React for Material Report
- Implemented Azure CI/CD pipelines with Docker, Helm, and ArgoCD for automated deployments to Kubernetes

Architech AI

June 2024 – Aug 2024

Software Engineer Intern | Python, PyTorch, OpenCV, Airflow, MongoDB

Taichung, Taiwan

- Resolved customer complaints about unwanted object generation by leading 2 developers to launch an object removal feature with Python, PyTorch and Stable Diffusion Model, resulting in **7.5%** of all feature usage
- Constructed a workflow with Python and PyTorch to segment areas based on free text, reducing process time by **50%** using Grounding DINO, SAM for text-driven image segmentation, and OpenCV for mask refinement
- Established an automated Airflow pipeline to provide plots of 6000+ data by room features to a Telegram bot

Intelligent Knowledge Management Lab

Sept 2022 – June 2023

Undergraduate Researcher | Python, PyTorch, Django

Tainan, Taiwan

- Deployed an AI-generated feedback loop to fine-tune a Large Language Model for code with 1 V100 GPU
- Utilized CodeGen-350M model to collect unsolved Mostly Basic Python Problems; achieved a **58%** pass@5 rate on original unsolved tasks during iterative feedback loop
- Developed a Django web application for users to compare original and fine-tuned models

Networked Artificial Intelligence Lab

Nov 2021 – Mar 2023

Research Assistant | Python, PyTorch, TensorFlow

Tainan, Taiwan

- Overcame imbalanced fake document detection by integrating SMOTE into Graph Neural Networks (GNN)
- Outperformed Text GNN model with a **15%** improvement in F1-score on Twitter bot and fake hotel reviews

PROJECTS

Fault-Tolerant Distributed File Systems

Nov 2025

- Engineered a Go Hadoop/Cassandra file system over RPC, ensuring 2-failure tolerance and eventual consistency
- Optimized merge performance by 25% via log compaction and skipping identical overwrites with sha256 hash
- Deployed on 10 VMs, using Bash to join/crash concurrently and Go to do distributed grep to trace concurrent logs

Intelligent Quality Assurance Tools

May 2025

- Built a VS Code extension with RAG workflow that using Gemini API to suggest test maintenance from code diffs
- Streamlined an AI workflow using Gemini API to parse Figma designs and generate BDD-style test cases

2nd Place, IEEE Big Data Cup: Trip Destination Prediction

Dec 2022

- Designed a model based on revisiting patterns and outperformed LightGBM and Random Forest
- Led a team to achieve 43% destination accuracy for over 1,000 trip destinations on nearly 1 million trips
- Performed exploratory data analysis using Seaborn and Matplotlib to identify correlations between key features

SKILLS

Languages: Python, C++, C#, Java, SQL, TypeScript, Go

Machine Learning: TensorFlow, PyTorch, Scikit-learn, pandas, OpenCV, Gemini API, Seaborn, Matplotlib

Cloud & DevOps: Azure, Docker, Kubernetes, Helm, Google Cloud Platform(GCP), Git, Airflow, Bash, RPC, TCP

Web & Databases: React, Node.js, Oracle SQL, MySQL, MongoDB, FastAPI